Problem Statement

Introducing a physical-validity metric for imputed vehicle state, and characterising the SFM percolation threshold in CACC-mixed traffic.

Domain

Kinematic feasibility of reconstructed trajectories; mixed-autonomy traffic simulation; vehicular network connectivity.

Motivation

Two gaps sit in the current evaluation. The evaluation reports positional accuracy only, with no measure of whether an imputed trajectory is kinematically possible — an interpolation method can score well on ADE while emitting states no vehicle could occupy. Separately, the claim that CAV-to-CAV leader connectivity undergoes a percolation-style jump rests on two data points: enough to say something happens between them, not enough to locate a threshold or rule out an artifact of how "leader" is defined. Neither task touches the downstream predictor, so both survive intact regardless of M1's verdict.

The Task

Define and compute an Infeasible Rate metric over existing pipeline outputs, sweep CAV penetration under multiple connectivity definitions, and implement a kinematic projection that enforces endpoint consistency across a gap.

Fixed parameters — authoritative, used from week 2 onward

These supersede all conflicting values elsewhere in the project documents.

Acceleration bounds: $[-8, +3]$ m/s²

Steering: ±45°

Wheelbase: $l_f, l_r \in [1, 2]$ m

Infeasible turning radius threshold: $r < 6$ m, where $r = 1/\kappa$

Simulation flow: 4500 veh/h, 1000 m 3-lane segment, speed limit 33.33 m/s

Module interface: kinematic_project(state_seq, dt, l_f, l_r) -> feasible_state_seq

Objectives

Infeasible Rate (mandatory, first). Curvature violation ($r < 6$ m) plus acceleration-bound violation per the Fixed parameters above, computed over the imputed windows the pipeline already produces, for all seven E1 conditions. Requires no new SUMO runs and no training.

Percolation sweep. Penetration at 18, 21, 24, 27, 30, 35% × 10 seeds (60 runs), fixed flow. Mean ± 95% CI, logistic fit, critical density and transition width. Adaptive top-up, once only: 10 additional seeds at the two levels bracketing the steepest slope. If CIs still overlap, report the transition as bounded between X and Y rather than located at a point. Min-gap secondary sweep dropped — flagged as future work.

Connectivity definitions — two. Modal leader over the window, and instantaneous per-timestep leader. These two answer artifact-vs-physics. The third (fraction of window with CAV leader) is added only if the first two disagree.

Kinematic projection. Method specified: forward integration with residual redistribution — integrate the bicycle model from the pre-gap state, compute terminal mismatch against the post-gap observation, distribute the correction across gap steps weighted by per-step feasible-control margin, verify no step violates the Fixed parameters after correction. Tested on a synthetic set of injected violations.

Mandatory core: Objective 1. Objectives 2–4 are extension work.

Deliverables
IR column added to Table 3, all seven conditions.
Percolation sweep figure with CIs, logistic fit, stated critical density or bounded interval.
Two-definition comparison with written artifact-vs-physics verdict.
kinematic_project module with tests.
"What we could not verify" section stating that Paper B's Tables II–III rest on the private CAERI-NDS dataset.
Stack
SUMO + TraCI (batch runner supplied by you), Python 3.10+, NumPy/SciPy, Matplotlib, Git.

DETAIL ABOUT:

This research assignment is based on the implementation and presentation of the problem statement assigned to each group.
This assignment will carry **25 marks**.

The last date of submission is **November 01, 2026, 11:59 pm**, through **Moodle only**.

We will do one Mid Sem evaluation also.

We will do a detailed presentation around the 3rd week of November 2026.

Expectation: Understand and implement the assigned problem statement (topic, problem, technology adopted, tool used).

Prepare a report (pdf) about your understanding, workflow, and the results that you generated.

You need to upload the source code to GitHub (with a proper readme) and share it with your mentor. Do not copy existing source code if it is already available.

You need to take approval for your code (it must be running) from your respective mentor.

Tools and Technologies

Your exact stack is listed in your assigned problem statement. Across this year's projects, the following tools and technologies are used, depending on the group:

- Python 3.10+, PyTorch, NumPy, Pandas, SciPy, Matplotlib, Git
- Docker and Docker Compose
- SUMO and TraCI, for vehicular and network simulation projects

- CloudSim Plus, for cloud scheduling and load balancing projects
- FastAPI or Flask, for building REST APIs
- nginx or Envoy, for traffic routing and load balancing
- Prometheus and Grafana, for monitoring and dashboards
- InfluxDB, for time series storage
- Apache JMeter, for load testing
- Kafka and MQTT (Mosquitto), for messaging
- SQLite, for local buffering and storage
- AWS EC2, or a free tier cloud GPU such as Google Colab or Kaggle, where specified
- Ollama or llama.cpp, ngrok, and Flower (flwr), for the specific LLM and federated learning projects that need them

If your problem statement lists a specific tool or version, follow that over this general list.

Marking: Implementation (12), Documentation and submission (5), Understanding and presentation (8).

Students must submit the assignment through Moodle on or before the given deadline. **Any one member can submit from each group**.

Submit in one master zip/tar/7z folder. Name your folder using the following convention:

<firstname>_<rollnumber>.zip (example: xyz_191CS101.zip)

The master folder must include 3 items:

- your project folder
- report (pdf)
- readme.txt